P8 / SAME-MATERIAL FINITE-ELEMENT STUDY / 26 OCT 2026

Six hardpoints. One PLA model.

Actual P6 geometry, complete printed normal-load paths, and two mounting assumptions. The initial 15 lb target is not imposed. The geometry has not been enlarged or quietly rounded to improve a stress plot.

Finding: N4 has the largest nominal normal-load stress screen with full rigid backing. With the two-hole fixture, N2 and N3 compare closely and N4 loses its lead. The mounted ranking is not stable enough to declare an actual maximum-weight winner. N5 needs a geometry revision.

How to read the numbers

These values are the first crossing of assigned local PLA stress thresholds in a linear FE model. They are **not measured breaking loads, nonlinear collapse loads, safe working loads or demonstrated release loads**. Several peaks depend on sharp CAD corners or ideal fixture edges. A physical maximum cannot honestly be inferred from an unconverged peak.

Design	Full rigid backing	Two-hole ideal fixture
N1 Straight keeper	4.24 N 0.43 kg / 0.95 lb	4.24 N 0.43 kg / 0.95 lb
N2 Supported cross-bolt	21.50 N 2.19 kg / 4.83 lb	11.26 N 1.15 kg / 2.53 lb
N3 Keyhole gate	14.49 N 1.48 kg / 3.26 lb	12.74 N 1.30 kg / 2.87 lb
N4 Side-exit T-rail	24.25 N 2.47 kg / 5.45 lb	10.50 N 1.07 kg / 2.36 lb
N5 Pivoting hook and sear	Unresolved contact See N5 branch cases	Unresolved contact See N5 branch cases
N6 Paired jaws and sear	8.41 N 0.86 kg / 1.89 lb	8.41 N 0.86 kg / 1.89 lb

Common h = 0.55 mm baseline, nominal selected contact cases, 1g static mass equivalent. N5 interior branch gives 0.713 N (0.07 kg); the alternate edge branch is hook-controlled at 2.04 N (0.21 kg). These branches are not physical confidence bounds. N6 two-hole values are mathematical fixture sensitivities: the current rear screw bore and washer seat are obstructed. All masses include departing adapter and hardware. No safety factor is applied.

For a specified resultant load factor n, divide the 1g mass equivalent by n. That conversion changes neither the contact assumptions nor the evidence level. Holding and successful separation remain different questions.

COMMON BASIS

Material, load and failure criterion

Input	Common value	Meaning
Elastic modulus	2,200 MPa	Effective homogenized PLA
Poisson ratio	0.35	Isotropic linear elasticity
Density	1.24 g/cm³	Bookkeeping; gravity represented by P
Tension / compression	25 / 40 MPa	Analyst-selected local thresholds
Maximum shear	15 MPa	Half principal stress difference
Print / environment	100% infill equivalent; 22°C	Short-duration static load; no sparse infill

The Bambu PLA Basic V3.0 sheet reports typical XY/Z moduli of 2,580/2,060 MPa and tensile strengths of 35/31 MPa for its 100% infill, annealed/dried coupons. This study selects the same effective values for every part, without favorable orientation credit. The compression and shear thresholds are assumptions, not values established by that sheet. [1]

The model does not resolve layer interfaces, road direction, voids, actual root radii, fracture, creep or hot-airframe exposure. Its 25 MPa tension threshold is not guaranteed conservative for a defective print. Bought screws and pivots remain metal; their force transfer is idealized. The solenoid is outside the intended retained transverse load path.

Stress-to-load calculation

$P = n\ m\ g$, with $g = 9.80665\ m/s^2$

$I = \max[\ \max(\sigma_1, 0)/25,\ \max(-\sigma_3, 0)/40,\ (\sigma_1 - \sigma_3)/(2 \times 15)\ ]$

$P_{cross} = 1 / \max(I)$, for a unit-load field

$m_{cross} = P_{cross} / (n\ g)$

σ_1 and σ_3 are the largest and smallest principal stress, in MPa. Each part is screened separately; the lowest crossing along the prescribed path controls. The N6 single-jaw submodel is solved at 1 N per jaw: equal sharing gives 2 × the jaw-unit threshold in total P, while one-jaw loading gives 1 ×.

Worked example, N3 adapter: h = 0.55 mm gives peak tensile stress 1.72475 MPa per N. Tension governs, so P = 25 / 1.72475 = 14.4949 N, equivalent to 1.4781 kg at 1g. At h = 0.4 mm it changes to 14.769 N. The two-hole mechanism is lower because the receiver fixture controls there.

[1] Bambu PLA Basic Technical Data Sheet V3.0. Analyst assumptions and complete source record: material.json.

MODELING CONTRACT

What was actually solved

Gmsh meshed the issued STEP solids. Scikit-fem solved 3D small-displacement elasticity with quadratic tetrahedral displacement elements on faceted boundaries. Finite pressure patches are imprinted without altering solid volume. Raw stress comes directly from the displacement gradient at element vertices, not smoothed nodal stress.

Boundary	Definition and practical limit
Full rigid backing	Receiver underside fixed in X/Y/Z. Represents an ideal continuously supported installation; an ordinary two-screw mounting does not automatically provide this.
Two-hole fixture	Available finite annular seating regions at the actual host holes fixed in X/Y/Z. It is a joint sensitivity, not a screw/washer/preload or aircraft-flexibility simulation.
Payload side	Complete outer adapter flange rigidly supported. Payload fastener pull-through and flange backing flexibility are not resolved.
Moving components	Prescribed balanced contact tractions; six scalar gauge constraints remove rigid modes. Their near-zero reactions demonstrate they are not carrying the intended load.
Contacts	Finite seated contact patterns with checked resultants and moments. They do not automatically find elastic load sharing, interference, sliding, contact take-up or loss of engagement.

Load paths

Design	Printed path included
N1 / N2	Adapter → keeper → receiver
N3	Adapter head → keyhole gate → receiver
N4	T-foot adapter → receiver rails; keeper has zero normal P
N5	Adapter → hook → pin/receiver and sear/receiver
N6	Adapter → one/two jaws → pins/receiver and sear/receiver

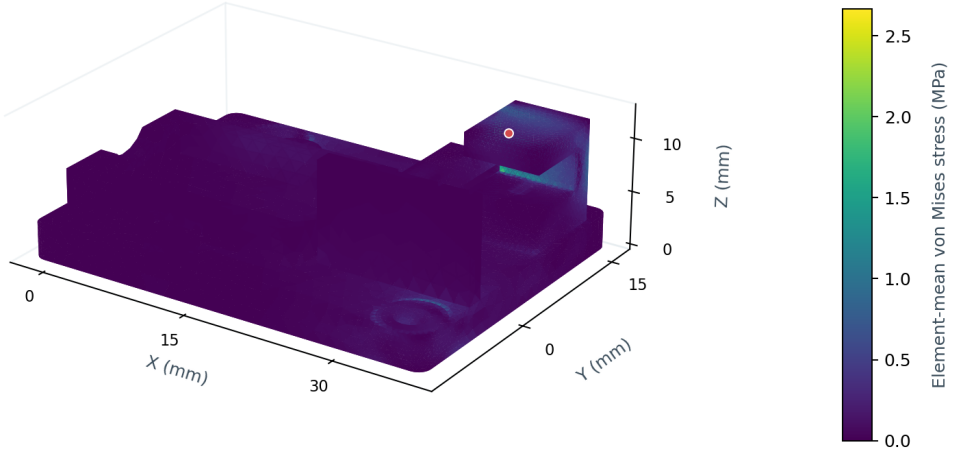
N6 left/right jaw geometry is exactly mirrored; the receiver is not symmetric, so both one-sided receiver cases are solved. N4 requires a separate longitudinal stop-load and departure calculation. None of this study demonstrates loaded release.

The rigid adapter fixture can react an eccentric contact moment. This is a stated fixture condition, not an assumption that every free-hanging payload has exactly the same force/moment wrench. Exact pressure centroids, faces, gauges, forces and moments are in each JSON input and result.

Straight keeper

N1 receiver · centered contact, ideal two-hole fixture

Actual STEP geometry · 1 N normal reference load · h=0.55 mm · 51,139 quadratic tetrahedra



Undeformed geometry. Color shows cell-mean stress for visualization, not the PLA failure criterion.
PLA comparison uses principal tension/compression and maximum shear. Red marker projects the raw hot-spot element centroid.
Raw peak tensile 5.894 MPa; compressive 3.863 MPa; shear 2.387 MPa at the unit load.
Prescribed finite contact tractions; ideal mounting. These are simulations, with no physical load rating.

Selected contact case	Full backing	Two-hole fixture
centered	4.237 N	4.242 N
narrow guide	4.516 N	4.522 N
low y half contact	3.887 N	3.891 N

R_{guide} = 2.06667 P; $R_{opposed}$ = 1.06667 P.

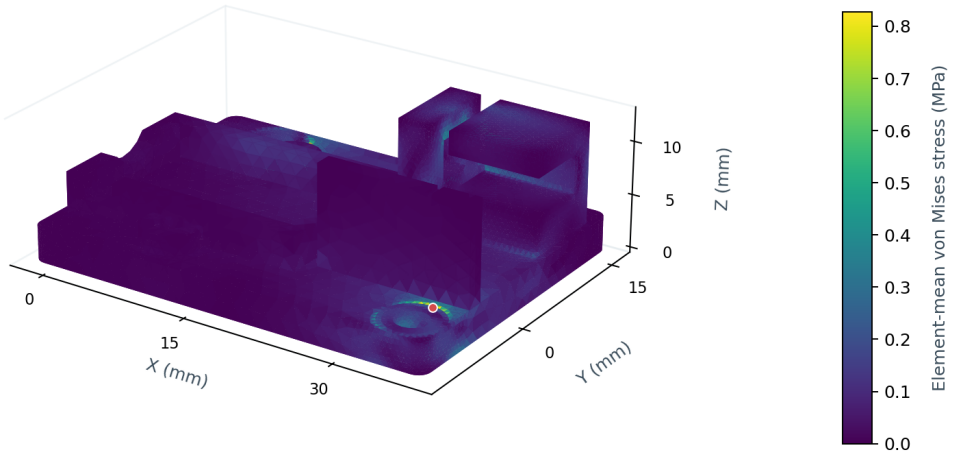
The receiver guide roof and its wall root govern before the keeper or adapter. At h = 0.55, the centered washer-fixture receiver moves about 0.0248 mm per N. A bounded local refinement changes its raw crossing from 4.24 to 4.34 N.

The opposed contact state requires clearance take-up, potentially about 0.15 rad in the nominal geometry. The solve does not establish that the actuator linkage allows this seating without side load. Thicken and round the guide root in a fit-checked revision before expecting much more capacity.

Supported cross-bolt

N2 receiver · both supports, ideal two-hole fixture

Actual STEP geometry · 1 N normal reference load · h=0.55 mm · 56,500 quadratic tetrahedra



Undeformed geometry. Color shows cell-mean stress for visualization, not the PLA failure criterion.
PLA comparison uses principal tension/compression and maximum shear. Red marker projects the raw hot-spot element centroid.
Raw peak tensile 1.272 MPa; compressive 3.551 MPa; shear 0.9581 MPa at the unit load.
Prescribed finite contact tractions; ideal mounting. These are simulations, with no physical load rating.

Selected contact case	Full backing	Two-hole fixture
centered	21.497 N	11.264 N
distal ineffective	7.483 N	7.527 N
low y half contact	17.817 N	11.578 N

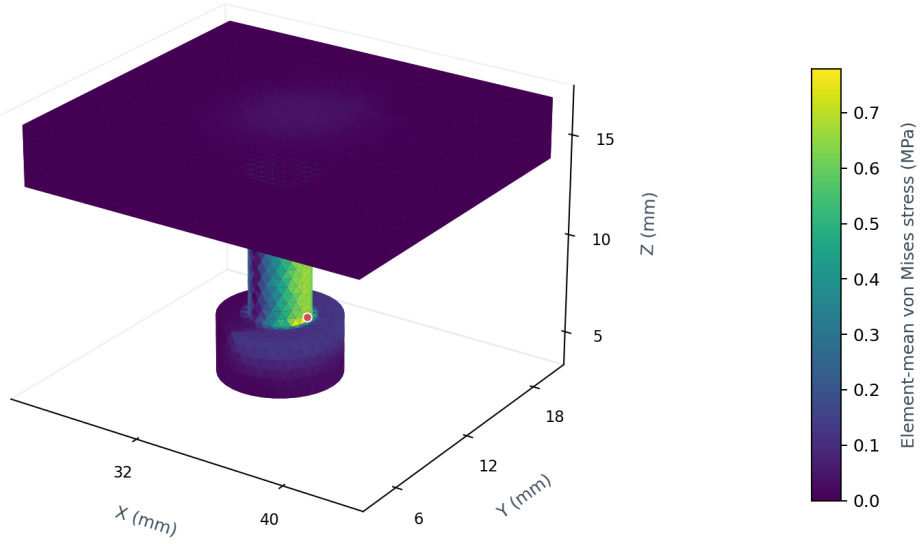
$R_{proximal} = 0.464286\ P$; $R_{distal} = 0.535714\ P$.

Supporting the cross-bolt on both sides improves the load path. Full backing gives 21.50 N in the nominal case; loss of the distal support reduces it to 7.48 N. With the washer fixture, the nominal raw peak is at an ideal annulus edge, while the lost-support case moves the concern back into the guide roof.

The extra local refinement gives 12.56 N nominal and 7.64 N with distal support ineffective for the washer fixture. The nominal last change remains about 11-12%; a finite physical peak is not established. Verify simultaneous support engagement and use a real backing/washer joint model for the installation.

Keyhole gate

N3 adapter · eccentric retained head contact
Actual STEP geometry · 1 N normal reference load · h=0.55 mm · 27,585 quadratic tetrahedra



Undeformed geometry. Color shows cell-mean stress for visualization, not the PLA failure criterion.
PLA comparison uses principal tension/compression and maximum shear. Red marker projects the raw hot-spot element centroid.
Raw peak tensile 1.725 MPa; compressive 0.7264 MPa; shear 0.6625 MPa at the unit load.
Prescribed finite contact tractions; ideal mounting. These are simulations, with no physical load rating.

Selected contact case	Full backing	Two-hole fixture
roof width 4	14.495 N	12.745 N
roof width 8	14.495 N	12.747 N

Head contact area = 7.55137 mm²; mean pressure = 0.13243 P MPa.
Pressure centroid is 1.48348 mm offset from the neck axis.

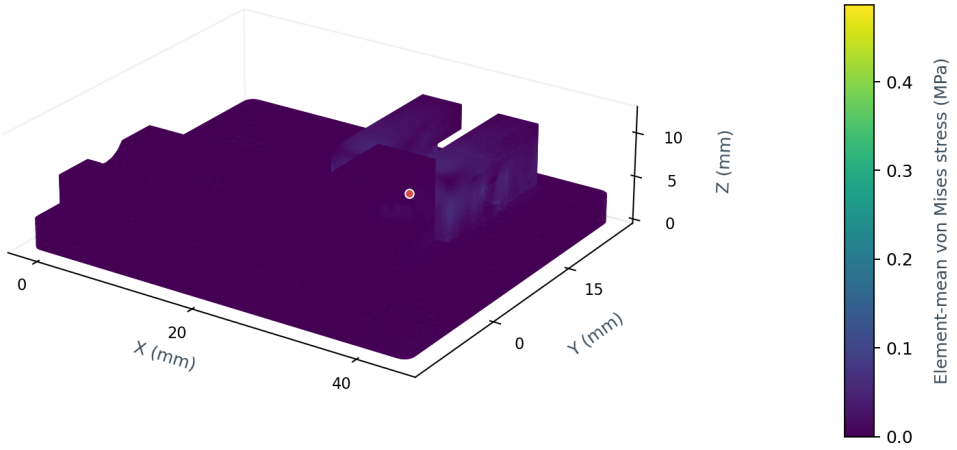
The sharp head/neck junction governs the fully backed path. The receiver fixture edge governs the two-hole path. Gate contact-width sensitivities of 4 and 8 mm do not change that mechanism-level conclusion. The adapter extra refinement changes 14.495 to 14.769 N, with stiffness nearly unchanged.

This is still the practical normal-departure candidate to refine and bench test. Add a deliberate manufacturable neck-root radius, enlarge bearing sections if needed, and verify gate clearance through the full stroke. The FEA does not verify the mini solenoid can withdraw the loaded gate.

Side-exit T-rail

N4 receiver · equal rails, full rigid backing

Actual STEP geometry · 1 N normal reference load · h=0.55 mm · 44,127 quadratic tetrahedra



Undeformed geometry. Color shows cell-mean stress for visualization, not the PLA failure criterion.
PLA comparison uses principal tension/compression and maximum shear. Red marker projects the raw hot-spot element centroid.
Raw peak tensile 1.031 MPa; compressive 0.4681 MPa; shear 0.4368 MPa at the unit load.
Prescribed finite contact tractions; ideal mounting. These are simulations, with no physical load rating.

Selected contact case	Full backing	Two-hole fixture
equal rails	24.248 N	10.498 N
left rail only	12.682 N	8.103 N
right rail only	12.050 N	13.514 N

Equal rail contact: P/2 per rail.
One-rail cases put the full P on a single rail and add an eccentric moment.

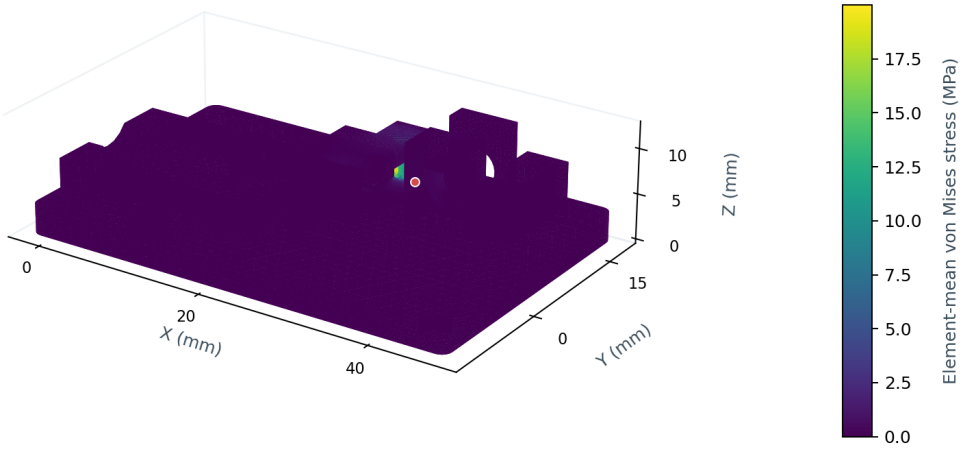
N4 gives the highest nominal fully backed screen: 24.25 N. Its rear-stop/lip junction, rather than the broad rail strip used in the earlier hand screen, concentrates stress. The two-hole nominal result falls to 10.50 N, and one-rail cases reduce the fully backed result to about 12 N.

The two-hole equal-rail raw crossing changes 18.72 → 15.02 → 10.50 N across the baseline meshes. This is not a converged strength. Real root geometry and a realistic host joint must be established. After unlocking, the adapter remains captured until a separate sideward force moves it out.

Pivoting hook and sear

N5 receiver · interior contact, full rigid backing

Actual STEP geometry · 1 N normal reference load · h=0.55 mm · 68,457 quadratic tetrahedra



Undeformed geometry. Color shows cell-mean stress for visualization, not the PLA failure criterion.
PLA comparison uses principal tension/compression and maximum shear. Red marker projects the raw hot-spot element centroid.
Raw peak tensile 35.09 MPa; compressive 43.94 MPa; shear 14.95 MPa at the unit load.
Prescribed finite contact tractions; ideal mounting. These are simulations, with no physical load rating.

Selected contact case	Full backing	Two-hole fixture
interior contact	0.713 N	0.712 N
edge contact	2.038 N	2.038 N

Sear load Q = 0.689655 P.
Interior roof branch: upward guide load = 0.448276 P.

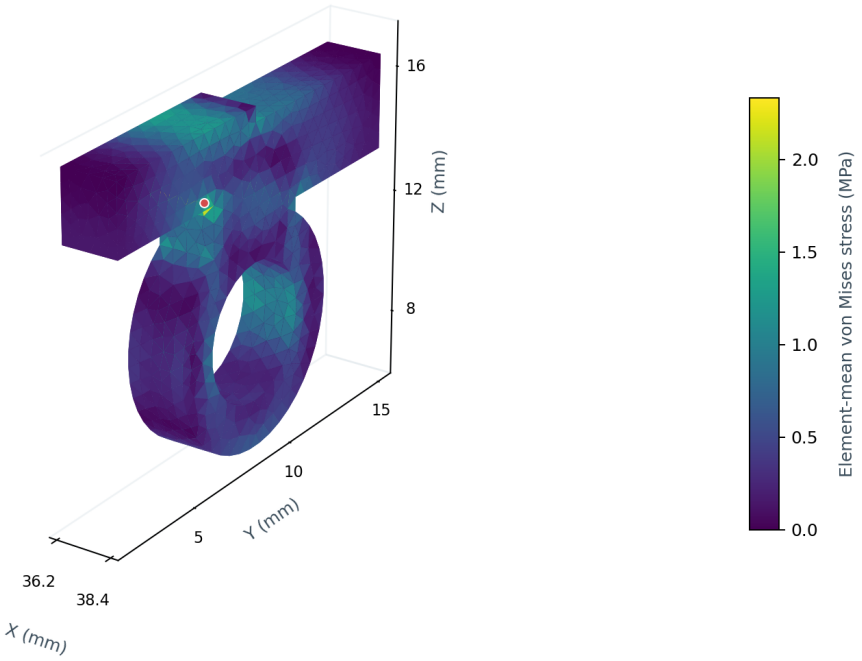
The flexible interior branch has a roughly 0.5 mm guide wall and about 1.81 mm displacement per N. Its raw local crossing is about 0.713 N, already far beyond clearance-scale motion. A distinct, stiff edge-contact branch is hook-controlled at 2.038 N.

The stiff branch relies on only 0.1 mm nominal roof overlap, less than the nominal lateral play. The simulation cannot select a unique assembled branch or extrapolate the flexible one to a real breaking weight. Revise the guide with positive overlap and continuous support, then rerun contact-aware analysis.

Paired jaws and sear

N6 jaw · 1 N per-jaw reference load

Actual STEP geometry · 1 N per-jaw reference load · h=0.55 mm · 5,952 quadratic tetrahedra



Undeformed geometry. Color shows cell-mean stress for visualization, not the PLA failure criterion.
PLA comparison uses principal tension/compression and maximum shear. Red marker projects the raw hot-spot element centroid.
Raw peak tensile 5.944 MPa; compressive 2.728 MPa; shear 2.037 MPa at the unit load.
Prescribed finite contact tractions; ideal mounting. These are simulations, with no physical load rating.

Selected contact case	Full backing	Two-hole fixture
equal jaws	8.412 N	8.412 N
right jaw only	4.206 N	4.206 N
left jaw only	4.206 N	4.206 N

$Q = 0.971963\ P_{jaw}$; pivot reaction $= 1.971963\ P_{jaw}$.
Equal: $P_{jaw} = P/2$. One-sided: $P_{jaw} = P$.

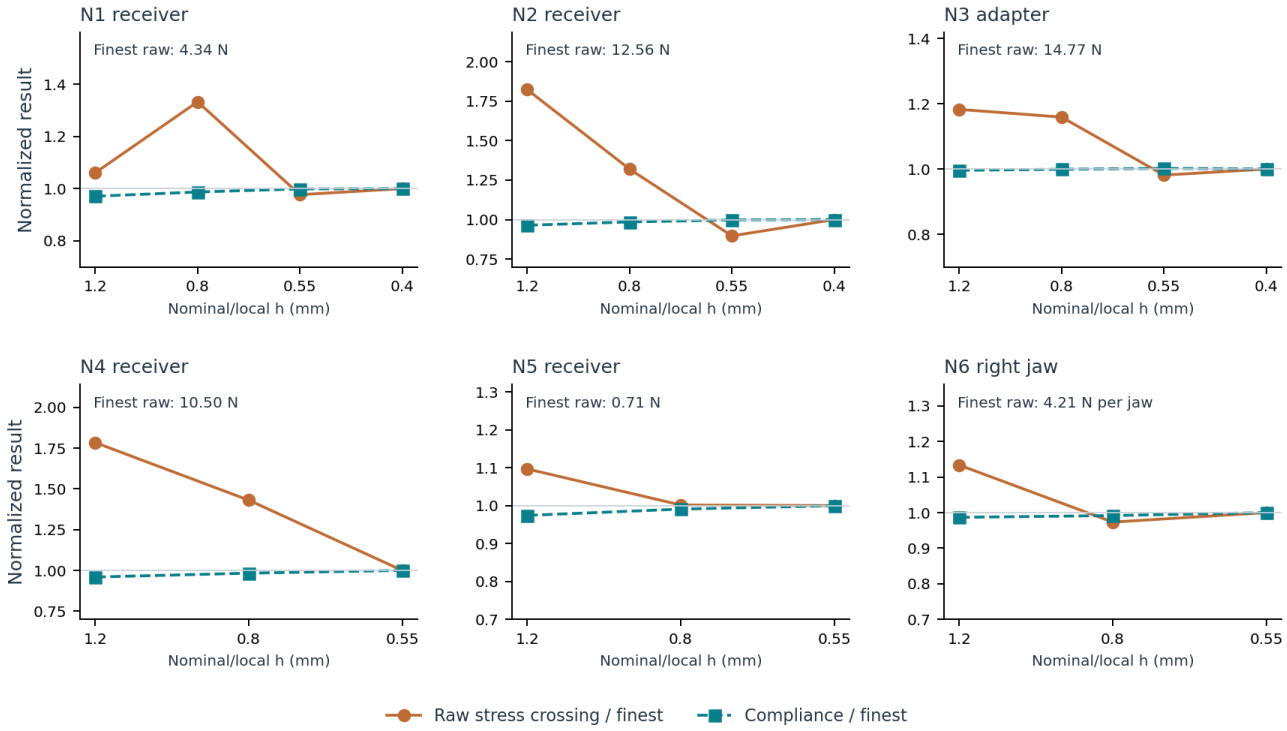
The two jaws only receive a factor-of-two credit when equal sharing exists. The jaw-unit raw crossing is 4.206 N at h = 0.55, or 8.412 N total under equal sharing. One-jaw contact removes that benefit. The sharp retaining-shelf/hub root is the principal jaw concern.

The receiver has asymmetric mounting geometry. Only 60.57% of the intended rear washer annulus is exposed; the nominal flat washer is obstructed by raised geometry. The rear guide also partly refills the screw bore: a nominal Ø3 mm shaft intersects 0.1513 mm³ of solid over Z = 3.1-3.2 mm. Revise bore clearance and the seat, and establish jaw equalization and pin/print fit before claiming the equal-share result.

CONVERGENCE / INTERPRETATION

A mesh-sensitive peak is not a load rating

Stable stiffness does not establish a finite peak-stress limit



N1/N2 h=0.4 is local refinement. Plot normalization is for convergence comparison only; it is not a capacity correction.

The baseline sizes are 1.2, 0.8 and 0.55 mm, with explicit local refinement boxes and coarser far fields. h is a mesher control, not every edge length. Additional h = 0.4 runs refine the N1/N2 receiver hotspots locally and the N3 adapter. All sequences are retained.

The reporting checks are less than 5% last-step compliance change and less than 10% raw-index change. Passing the last step alone does not establish an asymptotic finite strength at a mathematically sharp root. Earlier oscillation, contact assumptions and physical notch radius still matter.

The raw maxima can rise or move as the mesh changes. Adding an arbitrary fillet, discarding a hotspot or choosing a stress percentile would change the question. The issued CAD is therefore unchanged, and the raw stress screens remain explicitly conditional.

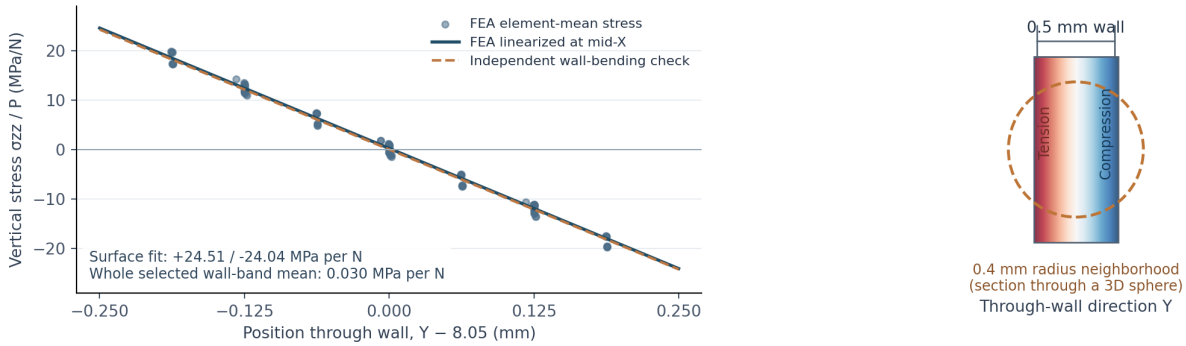
The output also contains tensor averages over 0.4 and 0.8 mm centroid neighborhoods. They are uncalibrated diagnostic sensitivities, **excluded from the capacity comparison**. Their length scale is not a measured PLA fracture property. In thin bending walls they can cancel opposing stresses and substantially overstate an apparent threshold.

N5 / INDEPENDENT CHECK

The thin-wall problem is real in this branch

N5 guide wall: opposite bending stresses can cancel in an average

Actual unchanged receiver • PLA E = 2200 MPa • 1 N prescribed load • h = 0.55 mm



The signed stress average is not a strength criterion. This prescribed-contact branch also becomes incompatible with simple clearance assumptions as it deflects.

Source: N5_receiver_bonded_fields.npz + receiver_wall_sanity.json; zero-radius source geometry and raw peaks remain in the results.

A simple wall calculation independently checks the finite-element result. For the interior contact branch, wall length is 3.8 mm, thickness 0.5 mm, roof lever 8.55 mm and upward roof load 0.448276 P.

$M = 0.448276\ P \times 8.55 = 3.83276\ P\ N{\cdot}mm$

$Z = 3.8 \times 0.5^2 / 6 = 0.15833\ mm^3$

$\sigma_{bending} = M/Z = 24.2069\ P\ MPa$

Estimated roof movement ≈ 1.5 P mm; FE maximum ≈ 1.81 P mm

An actual FE stress linearization through the midwall predicts +24.51 and −24.04 MPa per N at the surfaces, consistent with the beam result. The volume-average axial stress there is only 0.030 MPa per N because opposite bending stresses cancel. That small average is not evidence of a strong wall.

The flexible branch reaches motions comparable to the nominal 0.3 mm clearance at loads of order 0.2 N. This comparison does not solve the contact transition; it shows why linear extrapolation of that fixed branch cannot establish an actual maximum. The alternate edge branch is separately modeled, not automatically substituted.

Reproducible check: n5_n6/receiver_wall_sanity.py and receiver_wall_sanity.json. Actual field source and figure scripts are included.

VERIFICATION / EVIDENCE

Solved, audited and reproducible

Check	Recorded result / limit
Scope	184 case solves; 66 meshes; 19 unique STEP load-path solids plus mirrored jaw equivalence.
Force equilibrium	Maximum error 8.6e-11 N for the unit reference cases.
Moment equilibrium	Maximum error 2.25e-09 N·mm.
Linear solve	Maximum free relative residual 1.17e-09; 412,776 largest DOF count.
Rigid-mode gauges	Maximum gauge reaction norm 7.62e-12 N.
Geometry volume	Maximum absolute CAD-to-mesh volume error 0.301%.
Code checks	Uniform uniaxial stress 0.0625 MPa at 1 N; raw threshold 400 N. Cantilever finest deflection 0.0581955 mm vs Timoshenko 0.0586591 mm (0.79% difference).

Mesh shape quality and raw hotspot-cell quality are audited per mesh. N6 receiver initially crashed in the Netgen optimization stage. The issued N6 receiver sequence consistently uses HXT meshing with the standard Gmsh optimizer; the failed attempt and the recovery are documented. Solver equilibrium checks do not validate a physical contact assumption.

Independent reviews

Separate agents developed N1/N2, N3/N4 and N5/N6 boundary models. N1/N2 and N3/N4 reviewed each other, followed by an independent assembly aggregation and N5/N6 interpretation review. Their records retain limitations and rejected interpretations.

Provenance limitation

The driver gained a cylindrical-face selector filter and configurable mesh algorithm/optimizer while the study ran. Some historical driver hashes were captured from the file at result-write time, not an immutable executing snapshot. Exact old driver bytes were not retained. The frozen final solver is the authoritative replay source; the original hashes remain untouched. A bounded replay smoke check verifies the unaffected N3 gate path, not universal byte identity.

These numerical checks verify solution consistency. No physical part, coupon, loaded release, aircraft mount or flight test was performed in this task.

NEXT DESIGN STEP

Turn conditional screens into a measured capacity

Design	Recommended geometry / installation change
N1	Support and round the guide roof root; establish contact take-up without solenoid side load.
N2	Guarantee distal support engagement; define a real washer/backing joint.
N3	Add a fit-checked neck-root radius; verify gate travel and actual head-bearing contact.
N4	Revise the stop/lip junction and backing; establish one-rail tolerance and sideward exit drive.
N5	Replace the thin, interrupted guide wall and narrow edge contact with continuous positive support.
N6	Clear the obstructed rear screw bore and washer seat; thicken/round jaw retaining-shelf roots and establish equal sharing.

A focused bench program

1. Freeze the print and inspect it. Use one PLA batch, 100% infill, recorded orientation and slicer settings. Record measured wall thickness, layer height, root radius, guide clearance and print defects. Reject cracks, missing walls and incomplete bearing surfaces before loading.

2. Calibrate material and load paths. Test at least five matching tensile coupons in the weakest relevant orientation and representative bearing/root coupons. Print at least three assemblies per selected contact/mounting condition for initial screening. Record force and displacement, not only whether the part survives.

3. Test the actual fixture. Use the intended backing, screws, washers and adapter mount. Measure load sharing where two supports exist. Apply stepped normal pull with resolution suited to the low-force cases; record first crack, permanent set, loss of engagement and ultimate separation separately. Include both one-sided cases and intended CG moments.

4. Establish acceptance for a chosen application. Select mass and flight load envelope, then define proof factor, allowable permanent set, deflection clearance and test sample plan. A proposed preliminary gate is no crack or engagement loss at the selected proof load and less than 0.1 mm permanent set after unloading; revise that displacement limit if the mechanism clearances demand less. This is a proposed test criterion, not an issued rating.

5. Test release separately. After holding qualification, measure withdrawal force and full departure at load, misalignment and temperature. Repeat at low departing force. Successful unlocking alone does not establish separation, especially for the rail or a lifting payload.

Reproduce and inspect

Extract both ZIPs into the same parent folder. README.md and REPRODUCE.md describe the frozen inputs, complete meshes/fields, scalar CSVs, benchmarks, reviews and source. Use reproduce.py filters for a bounded replay into a new output directory. The website retains the six rotatable assemblies and adds the PLA study.